

Every Boundary F_σ Set Is a Banach Maximal Domain

Candidate full solution packet
arXiv:2407.15997, Problem 4.7

11 August 2026

Status. Candidate full affirmative answer in the Banach category, subject to expert review. For every F_σ set $\Gamma \subset \mathbb{T}$ we construct a Banach space of holomorphic functions on $\mathbb{D}$ satisfying P1–P3 of Mironov–Sampat and having maximal domain precisely $\mathbb{D} \cup \Gamma$.

1 Source problem

Let $\mathcal{X} \subset \text{Hol}(\mathbb{D})$ be a Hausdorff topological vector space such that:

- (P1) the analytic polynomials are dense in $\mathcal{X}$;
- (P2) evaluation at every point of $\mathbb{D}$ is continuous;
- (P3) multiplication by z is continuous on $\mathcal{X}$.

Its maximal domain is

$$\Omega_{\max}(\mathcal{X}) = \{w \in \mathbb{C} : p \mapsto p(w) \text{ is continuous on the polynomial subspace of } \mathcal{X}\}.$$

After realizing $\mathbb{D} \cup \Gamma$ for Γ both F_σ and G_δ , Mironov and Sampat ask the following on printed page 13 of [1].

Thus, (4.11) holds, and this completes the proof. ■

Remark 4.6. Examples of $\Gamma \subset \mathbb{T} \cong [0, 1)$ for which the above theorem works are: (1) any closed/open set, e.g., finite collection of closed/open sub-arcs of $\mathbb{T}$, the Cantor set and its complement, etc. (2) any countable union of intervals with countably many limit points in the complement, e.g., any sequence with countably many limit points, etc. Since our argument crucially relies on Γ also being G_δ , we do not know if our construction generalizes to, for instance, $\Gamma = \mathbb{Q} \cap [0, 1)$. We therefore ask the following question.

Problem 4.7. Does there exist a Hilbert/Banach space $\mathcal{H}$ satisfying **P1–P3** on $\mathbb{D}$ with $\Omega_{\max} = \mathbb{D} \cup \Gamma$ for any given F_σ set $\Gamma \subset \mathbb{T}$?

Remark 4.8. The above problem can be solved for locally convex TVSs using a similar idea as the previous theorems. We provide a sketch of the argument here. Suppose $\Gamma = \cup_k \Gamma_k$ with Γ_k closed, and consider the seminorms for $n, k \in \mathbb{N}$

$$p_n(f) = \sup_{z \in (\frac{n-1}{n})\mathbb{D}} |f(z)| \text{ and } q_k(f) = \sup_{z \in \Gamma_k} |f(z)|.$$

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The phrase “Hilbert/Banach” permits an affirmative Banach-space construction. We do not claim here that the space can always be Hilbert.

2 Proof intuition

Write Γ as a countable union of closed sets K_j . The natural plan is to impose a norm for each K_j which makes evaluation continuous on K_j , and then take a summable combination of those norms. The obstruction is uniformity: for a boundary peak at $w \notin \Gamma$, the distance from w to K_j may tend to zero arbitrarily quickly as j grows.

The key is a local norm whose peak estimates are bounded independently of that distance. Let $\mathcal{A}_0$ be the analytic coefficient space whose Taylor coefficients form a c_0 sequence. At $\gamma \in \mathbb{T}$, control

$$p(\gamma), \quad \Delta_\gamma p(z) := \frac{p(z) - p(\gamma)}{z - \gamma}$$

in $\mathbb{C} \oplus \mathcal{A}_0$. This is weak enough that the divided differences of the standard peaks are uniformly bounded. It is also strong enough to be an honest function norm: the only possible kernel would contain $1/(\gamma - z)$, whose Taylor coefficients have constant modulus and hence do not belong to c_0 .

Away from the peak, the divided-difference coefficients are oscillatory tails of a binomial distribution. Abel summation makes their supremum tend to zero. Thus every fixed closed layer contributes a quantity tending to zero, all layers have a common bound, and dominated convergence handles the summable outer norm. An H^2 term supplies interior evaluation and a canonical holomorphic limit; the local injectivity just described rules out invisible completion vectors.

3 The local coefficient model

Define

$$\mathcal{A}_0 = \left\{ g(z) = \sum_{n=0}^{\infty} a_n z^n : (a_n)_{n \geq 0} \in c_0 \right\}, \quad \|g\|_{\mathcal{A}_0} = \sup_{n \geq 0} |a_n|. \quad (1)$$

Every member of $\mathcal{A}_0$ is holomorphic on $\mathbb{D}$, convergence in $\mathcal{A}_0$ implies locally uniform convergence, polynomials are dense, and multiplication by z is an isometry.

For $\gamma \in \mathbb{T}$ and a polynomial p , put

$$\|p\|_\gamma := |p(\gamma)| + \|\Delta_\gamma p\|_{\mathcal{A}_0}. \quad (2)$$

Lemma 1 (No local ghost vectors). *The completion of the polynomials for (2) is isometric to $\mathbb{C} \oplus_1 \mathcal{A}_0$ through*

$$p \mapsto (p(\gamma), \Delta_\gamma p), \quad (3)$$

and the associated map into $\text{Hol}(\mathbb{D})$,

$$(c, g) \mapsto c + (z - \gamma)g(z), \quad (4)$$

is injective. Moreover,

$$\|zp\|_\gamma \leq 2\|p\|_\gamma \quad (p \in \mathcal{P}). \quad (5)$$

Proof. Finite Taylor truncations in $\mathcal{A}_0$ show that every pair (c, g) is a limit of polynomials $c + (z - \gamma)g_m$, so (3) has dense range. If $c + (z - \gamma)g(z) = 0$ on $\mathbb{D}$, then

$$g(z) = \frac{c}{\gamma - z} = \sum_{n=0}^{\infty} c\gamma^{-n-1}z^n.$$

The coefficients on the right have constant modulus $|c|$. Since the coefficients of g tend to zero, $c = 0$, and then $g = 0$. This proves injectivity.

Writing $p = c + (z - \gamma)g$ gives

$$zp = \gamma c + (z - \gamma)(c + zg).$$

The constant has $\mathcal{A}_0$ -norm $|c|$ and z is an isometry on $\mathcal{A}_0$; hence

$$\|zp\|_{\gamma} \leq |c| + |c| + \|g\|_{\mathcal{A}_0} \leq 2\|p\|_{\gamma}.$$

□

For a closed $K \subset \mathbb{T}$, with the empty supremum interpreted as zero, set

$$\|p\|_K := \sup_{\gamma \in K} \|p\|_{\gamma}. \quad (6)$$

4 Construction of the Banach function space

Fix an arbitrary F_{σ} set $\Gamma \subset \mathbb{T}$ and closed sets $K_j \subset \mathbb{T}$ such that

$$\Gamma = \bigcup_{j=1}^{\infty} K_j. \quad (7)$$

On the polynomials define

$$\|p\|_{\mathcal{X}} := \|p\|_{H^2} + \sum_{j=1}^{\infty} 2^{-j} \|p\|_{K_j}, \quad (8)$$

and let $\mathcal{X}$ be the abstract completion.

Proposition 2. *The natural map from $\mathcal{X}$ into $H^2 \subset \text{Hol}(\mathbb{D})$ is injective. Consequently $\mathcal{X}$ is a Banach space of holomorphic functions on $\mathbb{D}$. It satisfies P1–P3, and*

$$\mathbb{D} \cup \Gamma \subset \Omega_{\max}(\mathcal{X}). \quad (9)$$

Proof. The H^2 term gives a continuous map $J : \mathcal{X} \rightarrow H^2$. We prove that J has zero kernel. Let (p_m) be Cauchy for (8) and suppose $p_m \rightarrow 0$ in H^2 , hence locally uniformly on $\mathbb{D}$.

Fix j and $\gamma \in K_j$. The pairs $(p_m(\gamma), \Delta_{\gamma} p_m)$ are Cauchy in $\mathbb{C} \oplus_1 \mathcal{A}_0$. If their limit is (c_{γ}, g_{γ}) , the identity

$$p_m(z) = p_m(\gamma) + (z - \gamma)\Delta_{\gamma} p_m(z)$$

and local uniform convergence give $c_{\gamma} + (z - \gamma)g_{\gamma}(z) = 0$ on $\mathbb{D}$. Lemma 1 forces $(c_{\gamma}, g_{\gamma}) = 0$.

Because (p_m) is Cauchy uniformly over K_j , this pointwise conclusion is uniform: given $\varepsilon > 0$, choose m_0 such that $\|p_m - p_n\|_{K_j} < \varepsilon$ for $m, n \geq m_0$, and then let $n \rightarrow \infty$ in each local norm. Thus $\|p_m\|_{K_j} \leq \varepsilon$ for $m \geq m_0$. Finally, Cauchyness in the weighted sum and Fatou's lemma yield

$$\sum_j 2^{-j} \|p_m\|_{K_j} \leq \liminf_{n \rightarrow \infty} \sum_j 2^{-j} \|p_m - p_n\|_{K_j} \rightarrow 0.$$

Together with $\|p_m\|_{H^2} \rightarrow 0$, this proves $p_m \rightarrow 0$ in $\mathcal{X}$ and hence injectivity of J .

P1 holds by construction. Since (8) dominates the H^2 norm, interior evaluations are continuous, proving P2. Lemma 1, (6), and the H^2 shift isometry give $\|zp\|_{\mathcal{X}} \leq 2\|p\|_{\mathcal{X}}$, so the shift extends boundedly and P3 holds. Finally, if $\gamma \in K_j$, then

$$|p(\gamma)| \leq \|p\|_{K_j} \leq 2^j \|p\|_{\mathcal{X}}.$$

Thus every point of Γ , as well as every point of $\mathbb{D}$, belongs to the maximal domain. $\square$

5 Uniform boundary peaks

Lemma 3. *Fix $w \in \mathbb{T}$ and define*

$$q_N(z) = \left(\frac{1 + \bar{w}z}{2} \right)^N. \quad (10)$$

Then:

1. $q_N(w) = 1$ and $\|q_N\|_{H^2} \rightarrow 0$;
2. $\sup_{N \geq 1, \gamma \in \mathbb{T}} \|q_N\|_{\gamma} \leq 2$;
3. if $K \subset \mathbb{T}$ is closed and $w \notin K$, then $\|q_N\|_K \rightarrow 0$.

Proof. Write

$$q_N(z) = \sum_{k=0}^N a_{N,k} z^k, \quad a_{N,k} = 2^{-N} \binom{N}{k} \bar{w}^k.$$

The standard binomial identity gives

$$\|q_N\|_{H^2}^2 = 4^{-N} \sum_{k=0}^N \binom{N}{k}^2 = 4^{-N} \binom{2N}{N} \rightarrow 0. \quad (11)$$

Polynomial division gives

$$\Delta_{\gamma} q_N(z) = \sum_{n=0}^{N-1} b_{N,n}(\gamma) z^n, \quad b_{N,n}(\gamma) = \sum_{k=n+1}^N a_{N,k} \gamma^{k-1-n}. \quad (12)$$

Since $|a_{N,k}| = 2^{-N} \binom{N}{k}$ and those numbers sum to one,

$$|b_{N,n}(\gamma)| \leq 1. \quad (13)$$

Also $|q_N(\gamma)| \leq 1$, proving assertion 2.

For assertion 3 put $u = \bar{w}\gamma$. Up to a unimodular factor, $b_{N,n}(\gamma)$ is the binomial tail

$$\sum_{k=n+1}^N 2^{-N} \binom{N}{k} u^k. \quad (14)$$

Abel summation, splitting the unimodal binomial coefficients at their mode, shows that for $u \neq 1$

$$\sup_{0 \leq n < N} |b_{N,n}(\gamma)| \leq \frac{C}{|1 - u|} \max_{0 \leq k \leq N} 2^{-N} \binom{N}{k} \leq \frac{C'}{|w - \gamma| \sqrt{N+1}}. \quad (15)$$

For completeness, the first inequality uses only $|\sum_{r=s}^t u^r| \leq 2/|1-u|$ on each monotone half; the second is the elementary central-binomial estimate.

If K is closed and does not contain w , then $\delta = \text{dist}(w, K) > 0$. Equation (15) makes the $\mathcal{A}_0$ term in (2) tend to zero uniformly for $\gamma \in K$. Moreover,

$$\sup_{\gamma \in K} |q_N(\gamma)| \leq \left(\sup_{\gamma \in K} \left| \frac{1 + \bar{w}\gamma}{2} \right| \right)^N = \rho_K^N$$

with $\rho_K < 1$. Hence $\|q_N\|_K \rightarrow 0$. □

6 Main theorem

Theorem 4. *For every F_σ set $\Gamma \subset \mathbb{T}$, the Banach space $\mathcal{X}$ obtained from (8) satisfies P1–P3 on $\mathbb{D}$ and*

$$\Omega_{\max}(\mathcal{X}) = \mathbb{D} \cup \Gamma. \quad (16)$$

Therefore Problem 4.7 of [1] has an affirmative answer in the Banach category.

Proof. Proposition 2 gives the inclusion from right to left in (16). Let $w \in \mathbb{T} \setminus \Gamma$. Then $w \notin K_j$ for every j . Lemma 3 gives

$$\|q_N\|_{K_j} \rightarrow 0 \quad \text{for every fixed } j, \quad \|q_N\|_{K_j} \leq 2 \quad \text{for all } N, j.$$

Dominated convergence for the counting measure with masses 2^{-j} , along with (11), yields

$$\|q_N\|_{\mathcal{X}} = \|q_N\|_{H^2} + \sum_j 2^{-j} \|q_N\|_{K_j} \rightarrow 0.$$

But $q_N(w) = 1$, so polynomial evaluation at w is not continuous on $\mathcal{X}$.

It remains to exclude $|w| > 1$. Put $r_N(z) = (z/w)^N$. Then $r_N(w) = 1$, $\|r_N\|_{H^2} = |w|^{-N}$, and, for every $\gamma \in \mathbb{T}$,

$$|r_N(\gamma)| = |w|^{-N}, \quad \Delta_\gamma r_N(z) = w^{-N} \sum_{n=0}^{N-1} \gamma^{N-1-n} z^n.$$

Thus $\|r_N\|_\gamma = 2|w|^{-N}$ uniformly in γ , and $\|r_N\|_{\mathcal{X}} \leq 3|w|^{-N} \rightarrow 0$. Evaluation at w is again unbounded. This proves that no point outside $\mathbb{D} \cup \Gamma$ lies in $\Omega_{\max}(\mathcal{X})$ and completes the proof. □

7 Verification, scope, and novelty

The proof is noncomputational. A finite script supplied with the packet checks the polynomial-division formula (12) and the global coefficient bound (13); it is only a sanity check. The essential reviewer checks are the Abel estimate (15) and the no-ghost argument in Proposition 2.

The result settles the existence question because a Banach space is one of the two requested categories. It does not prove that every such maximal domain has a Hilbert realization. It also does not address the separate higher-dimensional question $\Omega_{\max} = \Omega_{\text{env}}$ in the source paper.

A bounded novelty search through 11 August 2026 covered the run indexes, arXiv:2407.15997v2, DOI 10.1090/proc/17344, exact title and problem wording, author and citation searches, and combinations of “maximal domain,” “ F_σ ,” “Hilbert/Banach,” and the rational-boundary example. The retrieved records still state only the source’s $F_\sigma \cap G_\delta$ Hilbert theorem, and no later answer to Problem 4.7 was found. This is not a proof of priority, so novelty confidence is moderate.

Human-review recommendation. Verify (15) uniformly in the tail index and the passage from pointwise local limits to the supremum and weighted-sum norms. Subject to those checks, the construction is a complete affirmative Banach-space solution.

References

- [1] M. Mironov and J. Sampat, *Jointly cyclic polynomials and maximal domains*, Proc. Amer. Math. Soc. **153** (2025), 4781–4795; arXiv:2407.15997v2, DOI: 10.1090/proc/17344.